

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

December 2016

EC203.01 Solid State Electronics

Date: 20.12.2016, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a)**
1. A semiconductor is formed by..... bonds. [01]
(i) covalent (ii) electrovalent (iii) co-ordinate (iv) none of the above
 2. The most commonly used semiconductor is [01]
(i) germanium (ii) silicon (iii) carbon (iv) sulphur
 3. A semiconductor has generally valence electrons. [01]
(i) 2 (ii) 3 (iii) 6 (iv) 4
 4. When a pure semiconductor is heated, its resistance [01]
(i) goes up (ii) goes down (iii) remains the same (iv) cannot say
 5. When a pentavalent impurity is added to a pure semiconductor, it becomes [01]
.....
(i) an insulator (ii) an intrinsic semiconductor (iii) p-type semiconductor (iv) n-type semiconductor
 6. Addition of pentavalent impurity to a semiconductor creates many [01]
(i) free electrons (ii) holes (iii) valence electrons (iv) bound electrons
 7. A hole in a semiconductor is defined as [01]
(i) a free electron (ii) the incomplete part of an electron pair bond (iii) a free proton (iv) a free neutron
 8. In a semiconductor, current conduction is due [01]
(i) only to holes (ii) only to free electrons (iii) to holes and free electrons (iv) none of the above
- Q - 2 (a)** Draw and explain circuit diagram of half wave rectifier. Derive its mathematical equation for maximum efficiency. [06]
- (b)** How half wave rectifier demerit overcome from centre tap full wave rectifier? Draw [07]
and explain circuit diagram of full wave rectifier using centre tap transformer. Derive

its mathematical equation for maximum efficiency.

OR

Q - 2 (a) A diode having internal resistance 5Ω is used for half wave rectification. If the applied voltage $35\sin \omega t$ and load resistance 800Ω . find [06]

i) I_m, I_{dc}, I_{rms}

ii) AC power input, DC power output

iii) DC output voltage

iv) Efficiency

(b) What is filter circuit in rectifier? Explain all types of filters use in rectifier with necessary circuit. [07]

Q - 3 (a) Draw and explain common emitter transistor configuration with its necessary input and output characteristics. [07]

(b) Explain the working of NPN transistor with necessary circuit diagram. [07]

OR

Q - 3 (a) What is stability factor in transistor? Derive mathematical expression of stability factor in transistor. [07]

(b) How transistor works as an amplifier? Describe it with necessary diagram. [07]

SECTION – II

Q - 4 (a) Draw and explain A.C. and D.C. equivalent circuit diagram of JFET amplifier. [07]

Q - 5 (a) Describe construction and working of junction field effect transistor with necessary diagram. [07]

(b) Compare BJT and JFET with its technical parameter. [07]

OR

Q - 5 (a) Describe construction and working of depletion mode metal oxide semiconductor field effect transistor (MOSFET) with necessary diagram. [07]

(b) Describe construction and working of enhancement mode metal oxide semiconductor field effect transistor (MOSFET) with necessary diagram. [07]

Q - 6 (a) Draw and explain circuit diagram of RC coupled multi stage amplifier. Enlist its advantages and disadvantages. [07]

(b) Draw and explain circuit diagram of transformer coupled multi stage amplifier. Enlist its advantages and disadvantages. [07]

OR

- Q - 6 (a)** Draw and explain circuit diagram of direct coupled multi stage amplifier. Enlist its advantages and disadvantages. **[07]**
- (b)** For given RC coupled two stage amplifiers, voltage divider bias resistor of two stages are $R_1=10K\Omega$, $R_2=5K\Omega$, $R_c=2K\Omega$, $R_E=2K\Omega$, $\beta=100$ and $V_{cc}=+15V$. If internal resistance of R_{in} of each stage is $1K\Omega$, determine **[07]**
- a) voltage gain of 1st stage
 - b) voltage gain of 2nd stage
 - c) Total voltage gain
